Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	6
Topic	Construction
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees
Theme	—
Curriculum objectives	• construct parallel lines • construct perpendicular lines • bisect a given line segment • construct angle 90° • construct angle 60°
Previous knowledge	• Students have been taught how to draw lines.
Lesson sequence	• construct parallel lines — the teacher draws the construction of parallel lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student construct parallel lines, following the teacher's board construction and replicating it in their notes • construct perpendicular lines — the teacher draws perpendicular lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student follow the teacher's board construction and replicate it in their notes • bisect a given line segment — the teacher draws the bisection of a line segment on the board step by step with a pair of compasses, explaining aloud as she draws; the student construct line segment, following the teacher's board construction and replicating it in their notes • construct angle 90° — the teacher constructs angle 90 degrees on the board step by step by compass arcs from a point on the line, explaining aloud as she constructs; the student construct angle 90 degrees, following the teacher's board construction and replicating it in their notes • construct angle 60° — the teacher constructs angle 60 degrees on the board step by step by compass arcs of equal radius from the endpoint, explaining aloud as she constructs; the student construct angle 60 degrees, following the teacher's board construction and replicating it in their notes
Instructional materials	• Mathematical set.
Evaluation	—

Lesson objectives

1.	construct parallel lines
2.	construct perpendicular lines
3.	bisect a given line segment
4.	construct angle 90°
5.	construct angle 60°

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

Class: JSS1 **Subject:** Mathematics **Week:** 6

Section 1: Constructing Parallel Lines

Parallel lines are lines in the same plane that never meet, however far they are extended. Two things you will use to draw them are a ruler and a set square (the triangle from your mathematical set).

Here is the idea behind the method: a set square has straight edges that meet at right angles. If you slide the set square along a straight edge without turning it, any line you draw along one of its edges will stay at the same angle to the edge you slid along. Lines drawn at the same angle to the same line are parallel to each other.

To draw a line parallel to a given line, follow these steps with your ruler and set square:

1. Draw your given line across your page with the ruler and label it (for example, call it line AB).
2. Place the ruler beside line AB so that one long edge of the ruler lies exactly along the line. Keep the ruler firmly in this position — it is your guide, and it must not move.
3. Rest one edge of the set square against the ruler's edge. The set square should touch the ruler so that its own edges make a right angle with the ruler's edge.
4. Holding the ruler still, slide the set square along the ruler to the place where you want the new parallel line to be.
5. Draw a straight line along the edge of the set square that is furthest from the ruler. That new line is parallel to your given line.

Because the set square slid without turning, its edge drew a line that makes exactly the same angle with the ruler as the given line does. Two lines that make the same angle with a third line are parallel, so your new line is parallel to line AB.

Worked example — draw a line parallel to line CD.

Suppose you have drawn line CD and you want a parallel line about 5 cm below it.

- Place your ruler so that one long edge lies exactly along line CD. Hold it there firmly.
- Set the set square against the ruler's edge, with its edges at a right angle to the ruler.
- Keeping the ruler still, slide the set square down the ruler until it is where you want the new line (about 5 cm below CD).
- Draw along the set square's free edge. Label the new line, for instance EF.

Line EF is parallel to line CD. If you slide the set square to several positions and draw each time, every line you draw will be parallel to CD — and therefore parallel to one another.

Practice

1. Using your ruler and set square, draw a line AB across your page. Then construct a second line parallel to AB, about 3 cm above it, and label it PQ.
2. Explain in your own words why, when you slide a set square along a ruler, every line you draw is parallel to the first line you drew.

Section 2: Constructing Perpendicular Lines

When two lines cross and the four angles where they meet are all right angles (each exactly 90°), the lines are called perpendicular. One quick way to recognise a pair of perpendicular lines is to look for a right-angle corner, like the corner of your set square, sitting neatly in the gap where the two lines meet.

You can construct a perpendicular line with just your set square and ruler. Here is the idea: the set square already has a built-in right angle, so if you line up one edge of that right angle with your given line and draw along the other edge, the new line you draw will meet the given line at a perfect right angle.

Follow these steps to draw a line perpendicular to a given line through a point on that line:

1. Draw a straight line and mark a point P on it. This point is where your perpendicular will cross the line.
2. Place your set square so that one edge of its right-angle corner lies exactly along the line, with the corner of the right angle sitting on point P.
3. Holding the set square steady, draw a straight line along the other edge of the right angle, starting from P.
4. The new line now meets the first line at P, and the corner where they meet is a right angle. Your two lines are perpendicular.

To check your work, slide the corner of the set square into the angle between the two lines. If the corner fits exactly with no gap, your lines are perpendicular.

Worked example: Construct a line perpendicular to AB at the point P on it.

Step 1. Draw a horizontal line and label its ends A and B. Mark a point P somewhere along AB.

Step 2. Take your set square and place it so that the edge of its right-angle corner lies along AB, with the corner of the right angle resting on point P.

Step 3. Keep the set square firmly in place and draw a straight line upward from P along the other edge of the right-angle corner.

Step 4. Check the corner: the angle between AB and your new line is a right angle, so the new line is perpendicular to AB through P.

Practice

1. Draw a horizontal line and mark a point Q on it. Using your set square and ruler, construct a line perpendicular to the horizontal line through Q. What should you check to confirm your construction is correct?
2. Two lines cross at a point, and when you check the corner with your set square it fits exactly with no gap. Are the two lines perpendicular? Explain why.

Section 3: Bisecting a Line Segment

To bisect a line segment means to cut it into two equal parts. When you bisect a segment, the point where the two equal parts meet is called the midpoint. The line that cuts the segment into equal halves crosses it at a right angle, so the construction also gives you a perpendicular through the middle of the segment.

You only need your mathematical set — a ruler and a pair of compasses. The compasses are the key tool here: instead of measuring the segment, you use equal radii (the opening of the compass) to locate the two points that sit exactly the same distance from each end.

Here is the plan you will follow every time:

Open your compasses to a radius greater than half the length of the segment. Keeping this radius fixed, place the point of the compasses on one end of the segment and draw an arc that passes above and below the segment.
Without changing the compass opening, move the point to the other end of the segment and draw another arc. The two arcs cross each other in two places — one above the segment and one below it.
Using your ruler, join those two crossing points with a straight line. This line passes through the midpoint of the segment and crosses it at a right angle.
The point where your new line meets the original segment is its midpoint, and it divides the segment into two equal parts.

The reason the arcs must be drawn with the same opening is that every point on an arc is the same distance from the compass point at the centre. Because you used the same radius from each end, each crossing point is the same distance from both ends — and a point that is equally far from the two ends lies exactly on the perpendicular bisector.

Worked example: Bisect a line segment AB that is about 8 cm long.

Draw a segment AB, about 8 cm long, with your ruler.
Open your compasses so the opening is a little more than half of AB — about 5 cm for an 8 cm segment.
Place the compass point on A. Draw an arc that curves above AB and another below it, using the same opening.
Move the compass point to B without changing the opening. Draw an arc above and below AB again. The new arcs cross the first pair in two points.
With your ruler, draw a straight line through the two crossing points. This line meets AB at its midpoint and stands at a right angle to it.
Label the midpoint M. You have now bisected AB: AM and MB are equal.

To check your work, measure AM and MB with the ruler. If you drew your arcs carefully with a fixed compass opening, the two lengths will be exactly equal.

Practice

1. Using only your ruler and compasses, bisect the line segment CD below (draw a segment of any convenient length first). Mark the midpoint and label it M, then write down the two equal parts it creates.
2. A builder wants to find the exact middle of a plank that is 120 cm long, without measuring it with a tape. Which instrument would she use to bisect the plank, and what is the length of each of the two equal parts?
3. When you bisect a line segment, you draw two arcs from each end of the segment. Why must the compass opening stay the same for all four arcs?

Section 4: Constructing an Angle of 90°

An angle of 90° is a right angle. Its two arms stand perpendicular to one another, like the corner of a book or the corner of this page. You can build one yourself with a compass and a ruler on any straight line.

Here is the method you will use:
1. Draw a straight line and mark a point on it where you want the 90° angle to sit.
2. Put the compass point on that point and draw an arc that cuts the line at two places, one on each side of your point.
3. Open the compass a little wider. Put the compass point on one of the two cuts and draw an arc above the line. Keep the opening the same, put the point on the other cut, and draw another arc above the line. The two arcs meet at a point above the line.
4. Draw a straight line from your marked point through that meeting point.
5. The angle between this new line and the original line is exactly 90°.

Let us construct a 90° angle at a point P on a straight line.

Step 1 — Draw a straight line and mark P somewhere along it.

Step 2 — Put the compass point on P and draw an arc that crosses the line on both sides of P. Call the two crossings A (on the left) and B (on the right).

Step 3 — Widen the compass a little. Put the point on A and draw an arc above the line. Keeping the compass exactly the same width, put the point on B and draw another arc above the line. The two arcs cross each other at a point C.

Step 4 — Use your ruler to draw a straight line from P up through C.

The angle between PC and the original line is 90°. Mark it with the ∟ symbol.

Practice

1. Using your ruler and compass, construct an angle of 90° on a straight line you draw yourself. Mark the point where the angle is made, and label the constructed right angle with the symbol ∟.

2. Look at this construction for an angle at point P: an arc from P cuts the line at A and B, and two wider arcs from A and B meet above the line at C. Without drawing anything else, what is the size of the angle APC?

Section 5: Constructing a 60° Angle with a Compass

A 60° angle is one of the angles you can build with just a straight edge and a compass — no protractor needed. The secret is that an equilateral triangle has three equal sides, and every one of its angles is exactly 60°. So if you draw an equilateral triangle, you have built a 60° angle for free.

You will need your mathematical set: a sharp pencil, a ruler (or straight edge), and a compass. Keep the compass setting fixed at the same radius through the whole construction — that is what makes the triangle equilateral.

Here is the whole construction in words before you try it yourself.

1. Draw a straight line and mark a point on it. Call that point A. This will be the vertex of your angle.
2. Set your compass to a comfortable radius. With the compass point on A, draw an arc that crosses the line. Call the point where the arc meets the line B.
3. Keep the compass at exactly the same radius. Now put the compass point on B and draw a second arc that cuts the first arc. Call the point where the two arcs meet C.
4. Use your ruler to join C to A with a straight line.

The line AC makes an angle of 60° with the line AB at A. Done — you have constructed a 60° angle.

Let's build a 60° angle together, step by step.

Start: Draw a line and mark point A on it. Set your compass to a radius of, say, 4 cm — any radius will work, as long as you never change it.

Step 1 — Draw the first arc. With the compass point on A, draw an arc that crosses the line. The point where the arc meets the line is B.

Step 2 — Draw the second arc. Without changing the compass radius, place the point on B and draw another arc so that it crosses the first arc. Where the two arcs meet, mark point C.

Step 3 — Join C to A. Use your ruler to draw the straight line from C back to A.

Now look at triangle ABC. The two arcs used the same radius, so AB = AC = BC. That makes triangle ABC equilateral, and every angle of an equilateral triangle is 60°. So the angle at A, between line AB and line AC, is exactly 60°. The construction is complete.

Practice

1. Using only your ruler and compass, construct a 60° angle. Follow the construction: draw a line, mark point A, set a compass radius, draw a first arc to meet the line at B, keep the same radius, draw a second arc from B to cut the first at C, then join C to A. When you finish, measure your angle with a protractor to check it is about 60°.
2. Why must you keep the compass at the same radius when you draw the second arc from point B?

Check your work

Attempt every question before you read this page.

Constructing Parallel Lines

1. The line PQ you drew should be parallel to AB: lay your ruler along AB, slide the set square along it (without turning it) to about 3 cm above, and draw along the set square's free edge.
   a. Draw line AB with the ruler.
   b. Place the ruler's long edge exactly along AB and hold it still.
   c. Rest the set square against the ruler's edge so its edges meet the ruler at a right angle.
   d. Slide the set square up the ruler to about 3 cm above AB, keeping the ruler fixed.

e. Draw along the edge of the set square furthest from the ruler and label it PQ. Since the set square never turned, PQ makes the same angle with the ruler as AB does, so PQ is parallel to AB.

2. Because the set square slides without turning, its drawing edge keeps the same angle to the ruler all the way along. Every line drawn along that edge therefore makes the same angle with the ruler as the first line, and lines that make the same angle with a third line are parallel.

Constructing Perpendicular Lines

1. You draw the new line through Q along the other edge of the set square's right-angle corner, with the corner resting on Q and one edge lying on the horizontal line. To check it, slide the corner of the set square into the angle where the two lines meet — if it fits exactly with no gap, your construction is correct because the lines meet at a right angle.

   Line the set square's right-angle edge along the horizontal line with the corner on Q, then draw upward along the other right-angle edge. The two lines meet at Q at 90°, so they are perpendicular.
2. Yes. The set square's corner is a right angle (90°). If it fits exactly in the angle where the lines meet, that angle is 90°, and two lines that meet at a right angle are perpendicular.

Bisecting a Line Segment

1. The construction produces a perpendicular line through the midpoint M of CD, so the segment is cut into two equal parts CM and MD, with CM = MD.

   Draw CD. Open the compasses wider than half of CD, draw an arc above and below CD from point C, then with the same opening draw arcs from point D. Join the two crossing points with a straight line. Where this line meets CD is the midpoint M, dividing the segment into two equal halves CM and MD.
2. She would use a pair of compasses and a ruler (her mathematical set) to bisect the plank. Each equal part is 120 cm ÷ 2 = 60 cm.

   Bisecting means cutting into two equal parts. Half of 120 cm is 60 cm, and bisection finds that exact middle point.
3. The same opening keeps the radius equal from both ends, so each crossing point of the arcs is the same distance from both ends. That places the crossing points exactly on the perpendicular bisector, which passes through the midpoint and cuts the segment into two equal parts.

   Every point on an arc is the same distance from the compass point. Using one fixed radius from both ends makes each intersection equally far from both A and B, so the line through the intersections is the perpendicular bisector that lands exactly on the midpoint.

Constructing an Angle of 90°

1. Your construction should show a point on a line, an arc cutting the line on both sides of that point, two wider arcs meeting above the line at a single crossing, and a straight line from the point up through that crossing. The angle between the new line and the original line is exactly 90°, marked with the ∟ symbol.
   a. Draw a straight line and mark a point P on it.
   b. With the compass point on P, draw an arc cutting the line at A (left) and B (right).
   c. Widen the compass. From A draw an arc above the line; keep the width fixed, from B draw another arc above the line so the two arcs meet at C.
   d. Draw the straight line from P through C. The angle between PC and the original line is 90°, so mark it with ∟.
2. The angle APC is 90°.

   The two wider arcs from A and B meet at C, and the line from P through C was drawn through that meeting point. This is exactly the construction for a right angle, so angle APC measures 90°.

Constructing a 60° Angle with a Compass

1. Your constructed angle at A should measure approximately 60°. It is exactly 60° if you kept the compass radius fixed throughout, because triangle ABC is then equilateral with all its sides equal, and every angle of an equilateral triangle is 60°.

   Draw line AB, set the compass to a fixed radius, draw an arc from A to cross the line at B, then from B draw another arc of the same radius to cut the first at C. Join C to A. Because both arcs used the same radius, AB = AC = BC, so triangle ABC is equilateral and angle BAC = 60°.
2. Keeping the same radius makes AC equal to AB and equal to BC, so triangle ABC has three equal sides. Only then is it an equilateral triangle, and only an equilateral triangle has all angles equal to 60°. If you changed the radius, the two sides would not be equal and the angle at A would not be 60°.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-14ba01e9	**Date**	
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____